INCH-POUND

MIL-DTL-23806/1D
w/AMENDMENT 1
13 January 2022
SUPERSEDING
MIL-DTL-23806/1D
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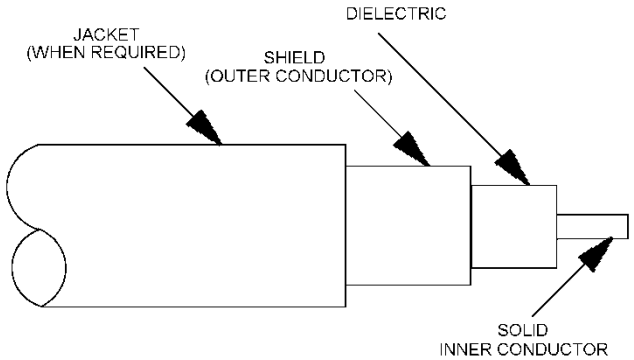
DETAIL SPECIFICATION SHEET

CABLE, RADIO FREQUENCY, COAXIAL, SEMIRIGID, FOAM DIELECTRIC, .500 INCH, 50 AND 75 OHM, (RG-231A/U, RG-331/U, RG-334/U AND RG-335/U)

Inactive for new design after 10 July 2000

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-23806.



Inches	mm
.001	0.03
.002	0.05
.004	0.10
.005	0.13
.098	2.49
.162	4.11
.450	11.43
.500	12.70
.625	15.88
5.	127.

Characteristics	50 ohms		75 ohms	
	Type I RG-231A/U	Type II RG-331/U	Type I RG-334/U	Type II RG-335/U
	Construction details			
Inner conductor				
Outside diameter	0.162 ± .002	0.162 ± .002	0.098 ± .001	0.098 ± .001
Outer conductor				
Inside diameter	0.450 ± .004	0.450 ± .004	0.450 ± .004	0.450 ± .004
Outside diameter	0.500 ± .005	0.500 ± .005	0.500 ± .005	0.500 ± .005
Jacket				
Minimum wall thickness		0.045		0.045
Maximum outside diameter		0.625		0.625
Minimum bending radius	5	5		
Weight (pounds per 1000 feet, maximum)	152	187	104	134

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are for information only.

FIGURE 1. Construction.

AMSC N/A FSC 6145

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ENGINEERING INFORMATION:

Continuous working voltage:	RG-231A/U RG-331/U	RG-334/U RG-335/U
	1,960 V_{rms}, max.	2,400 V_{rms}, max.

Frequency range: 10 GHz, maximum.
Power rating: See figure 2.
Operating temperature range: -55 °C to +85 °C.

REQUIREMENTS (RG-231A/U and RG-331/U):

Design and construction: See figure 1.
Impedance: 50 $\pm$ 1.
Capacitance: 25 pF/ft, nominal.
Velocity of propagation: 81 $\pm$ 2 percent.
Dielectric strength (at 60 Hz): 3,500 V_{rms}, minimum.
Dielectric constant: 1.5, nominal.
Attenuation: See figure 2.

REQUIREMENTS (RG-334/U and RG-335/U):

Design and construction: See figure 1.
Impedance: 75 $\pm$ 2.
Capacitance: 17 pF/ft, nominal.
Velocity of propagation: 81 $\pm$ 2 percent.
Dielectric strength (at 60 Hz): 4,300 V_{rms}, minimum.
Dielectric constant: 1.5, nominal.
Attenuation: See figure 2.

REQUIREMENTS (All cables):

The attenuation measurement for all cables is tested at 30 MHz, 400 MHz and 3 GHz; maximum attenuation is not to exceed 0.53 dB, 2.19 dB and 7.0 dB per 100 feet.

Voltage standing wave ratio (VSWR): See table I.

TABLE I. Voltage standing wave ratio.

Frequency range (MHz)	Initial (maximum)	After temperature cycling (maximum)	After cooling (maximum)
500 to 2000	1.15	1.18	1.18
2000 to 5000	1.20	1.25	1.25
5000 to 10,000	1.30	1.35	1.35

Temperature cycling and bending test: Coiling mandrel, 15 inches in diameter, maximum.

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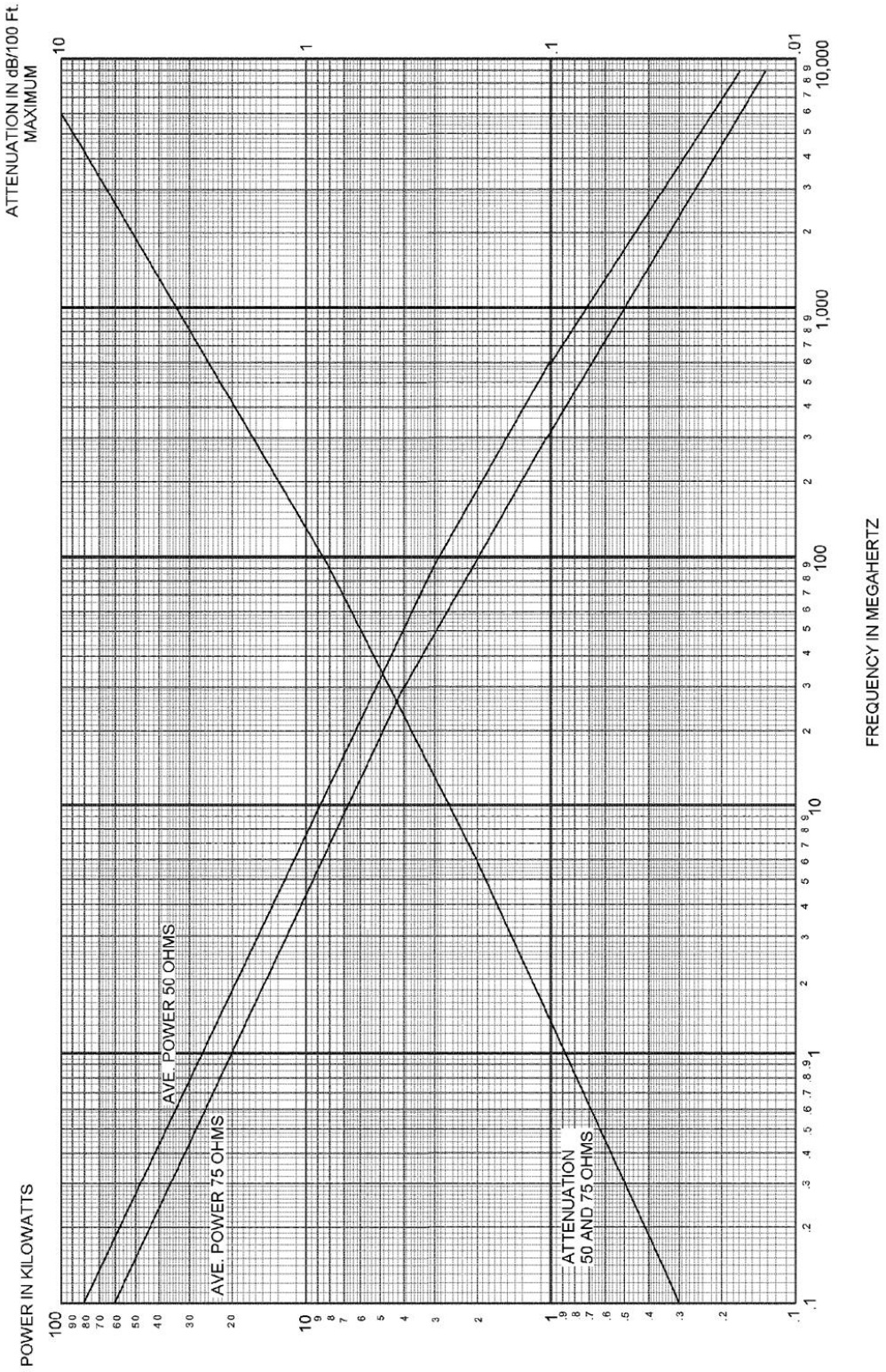


FIGURE 2. .500 inch foam dielectric coaxial cable (50 and 75 ohm).

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Amendment Notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Referenced documents. This document references MIL-DTL-23806.

CONCLUDING MATERIAL

Custodians:
Army – CR
Navy – EC
Air Force – 85
DLA – CC

Preparing activity:
DLA – CC

(Project 6145-2022-002)

Review activities:
Army – AR, AV, MI
Navy – MC, MS, SA, YD
Air Force – 71

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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